

Music and AI

Image and Video Analysis and Technology

Spyros Kantarelis, 25/5/2026

AI | LS

What is Music?

Sound vs Music

Sound is **vibrations** that travel through the air or another medium and can be heard when they reach a person's or animal's ear.

Humans can detect sounds in a frequency range from about **20 Hz to 20 kHz**.

Music is the **artful arrangement of silence and sound**.

Music is crafted to **weave narratives**.

Music expresses and evokes **emotions and thoughts**.

Music is a **universal language**.

What is Music?

Music as a Language

	Natural Language	Music
Foundation Blocks	Letters → Words → Sentences → Paragraphs	Notes → Phrases → Sections → Compositions
Syntax	Word order, punctuation, sentence structure	Chord progressions, cadences, rhythmic patterns
Vocabulary	Dictionary of words with defined meanings	Scales, chords, intervals, dynamics
Expression	Tone, emphasis, pacing	Timbre, articulation, tempo

Music and Computer Science

Music Information Retrieval (MIR)

MIR is **the interdisciplinary science** of retrieving information from music.

Core Disciplines: Computer Science and AI, Signal Processing, Musicology, Psychology, Data Science

Main Tasks: Music Generation, Music Classification, Music Analysis, Music Transcription, Music Recommendation, Source Separation

Main Conferences: International Society for Music Information Retrieval (ISMIR), International Conference on Acoustics, Speech, and Signal Processing (ICASSP)

Music and Computer Science

Music Information Retrieval (MIR)

The MIREX¹ (Music Information Retrieval Evaluation eXchange) topics:

- [2005:Audio Artist Identification](#) - Deadline: 3 August 2005
- [2005:Audio Drum Detection](#) - Deadline: Wednesday, 20 July 2005
- [2005:Audio Genre Classification](#) - Deadline: 2 August 2005
- [2005:Audio Melody Extraction](#) - Deadline: Tuesday, 19 July 2005
- [2005:Audio Onset Detection](#) - Deadline: 1 August 2005
- [2005:Audio Tempo Extraction](#) - Deadline: Monday, 18 July 2005
- [2005:Audio and Symbolic Key Finding](#) - Deadline: 1 August 2005
- [2005:Symbolic Genre Classification](#) - Deadline: 1 August 2005
- [2005:Symbolic Melodic Similarity](#) - Deadline: 3 August 2005

MIREX topics in 2005

1. https://www.music-ir.org/mirex/wiki/MIREX_HOME

Music and Computer Science

Music Information Retrieval (MIR)

The MIREX¹ (Music Information Retrieval Evaluation eXchange) topics:

- 2005:Audio Artist Identification - Deadline: 3 August 2005
- 2005:Audio Drum Detection - Deadline: Wednesday, 20 July 2005
- 2005:Audio Genre Classification - Deadline: 2 August 2005
- 2005:Audio Melody Extraction - Deadline: Tuesday, 19 July 2005
- 2005:Audio Onset Detection - Deadline: 1 August 2005
- 2005:Audio Tempo Extraction - Deadline: Monday, 18 July 2005
- 2005:Audio and Symbolic Key Finding - Deadline: 1 August 2005
- 2005:Symbolic Genre Classification - Deadline: 1 August 2005
- 2005:Symbolic Melodic Similarity - Deadline: 3 August 2005

MIREX topics in 2005



- Traditional MIR tasks
- 2025:Audio Chord Estimation <TC: Junyan Jiang >
 - 2025:Lyrics Transcription <TC: Ruibin Yuan  & Junyan Jiang >
 - 2025:Cover Song Identification <TC: Xingjian Du  & Ruibin Yuan >
 - 2025:Music Structure Analysis <TC: Yixiao Zhang  & Ruibin Yuan >
 - 2025:Audio Beat Tracking <TC: Wenye Ma  & Yinghao Ma >
 - 2025:Audio Key Detection <TC: Wenye Ma  & Yinghao Ma >
- Modern MIR Tasks
- 2025:Symbolic Music Generation <TC: Ziyu Wang  & Jingwei Zhao >
 - 2025:Music Audio Generation <TC: Ruibin Yuan >
 - 2025:Music Description & Captioning <TC: Yixiao Zhang  & Ruibin Yuan >
 - 2025:Polyphonic Transcription <TC: Ojas Chaturvedi , Yung-Hsiang Lu , K
 - 2025:Song Deepfake Detection <TC: Neil Zhang >
 - 2025:Music Reasoning QA <TC: Yinghao Ma >
 - 2025:RenCon (Expressive Piano Performance Rendering Competition) <TC: H

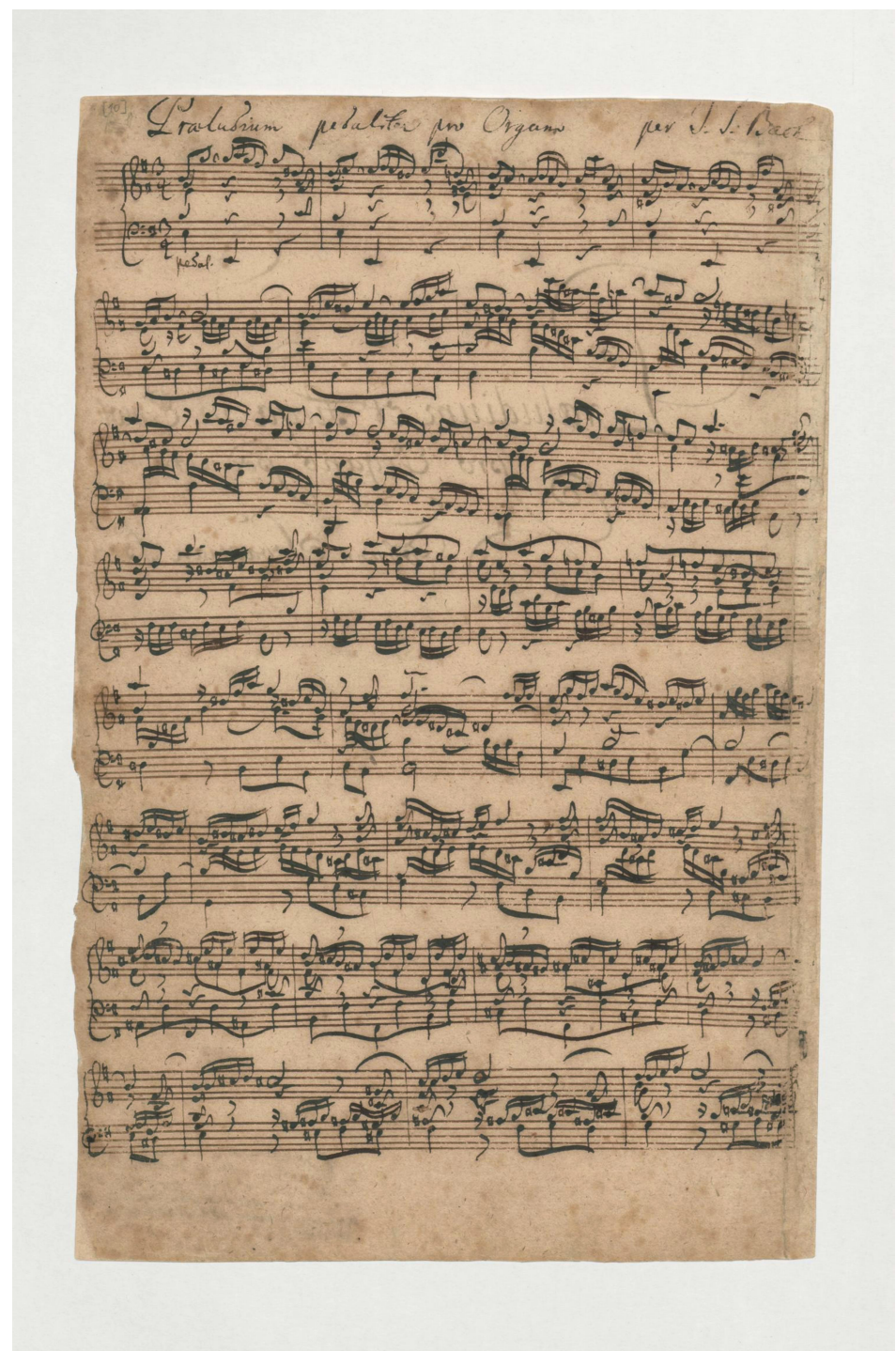
MIREX topics in 2025

1. https://www.music-ir.org/mirex/wiki/MIREX_HOME

Symbolic Music

Symbolic Music Representation

Symbolic music offers **structured information**, allowing the analysis of musical elements independent of audio recordings.



Sheet

(three-phase binary form)
Sarabande
(establishment of tonic)

(move to relative major)

(move to subdominant)

Digital Sheet

Time (Ticks)	Message	Channel	Note Number	Velocity
60	NOTE ON	1	67	100
0	NOTE ON	1	55	100
0	NOTE ON	2	43	100
55	NOTE OFF	1	67	0
0	NOTE OFF	1	55	0
0	NOTE OFF	2	43	0
5	NOTE ON	1	67	100
0	NOTE ON	1	55	100
0	NOTE ON	2	43	100
55	NOTE OFF	1	67	0
0	NOTE OFF	1	55	0
0	NOTE OFF	2	43	0
5	NOTE ON	1	67	100
0	NOTE ON	1	55	100
0	NOTE ON	2	43	100
55	NOTE OFF	1	67	0
0	NOTE OFF	1	55	0
0	NOTE OFF	2	43	0
5	NOTE ON	1	63	100
0	NOTE ON	2	51	100
0	NOTE ON	2	39	100
240	NOTE OFF	1	63	0
0	NOTE OFF	2	51	0
0	NOTE OFF	2	39	0

MIDI messages

C - G - Am - F - E

I - V - vi - IV - III

Text-based



Symbolic Music

Basic Tools

- MusicXML - <https://www.musicxml.com/>
- Mido (python library) - <https://mido.readthedocs.io/en/stable/>
- ChordAI - <https://chordai.net/>
- Hookpad - <https://www.hooktheory.com/hookpad>
- music21 (python library) - <https://www.music21.org/music21docs/>
- jSymbolic - <https://jmir.sourceforge.net/jSymbolic.html>

Symbolic Music

Main Tasks

- Symbolic Music Generation
- Symbolic Music Analysis
- Symbolic Music Standardisation and Integration
- Feature Extraction
- Pattern Discovery

Symbolic Music

Semantic Web and Description Logics

Using Semantic Web technologies and Description Logics, we can enrich music data with **machine-interpretable metadata** and **extract meaningful features** through Knowledge.

Here is an example triple using RDF and the *Music Ontology* domain:

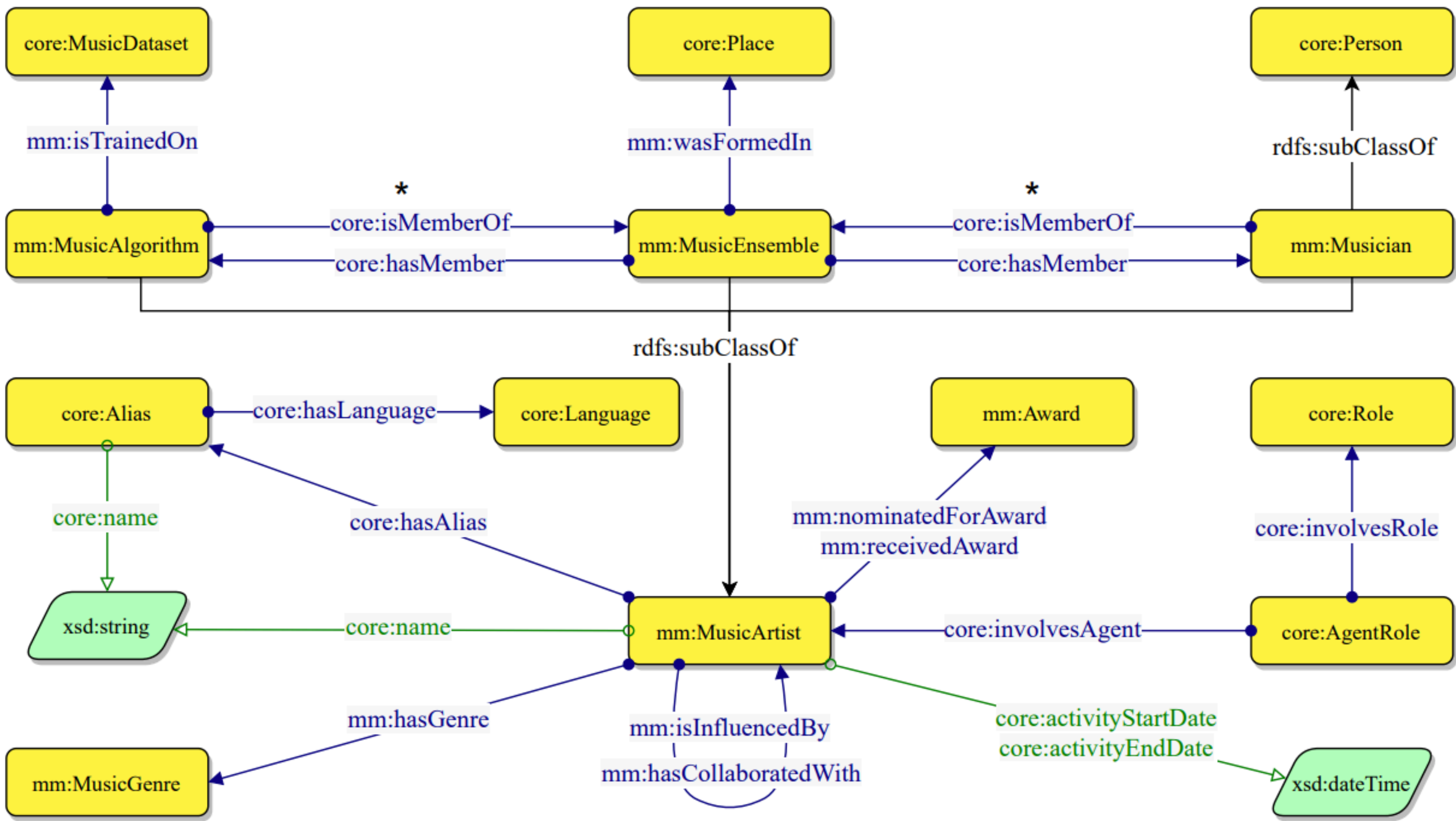
`<http://example.org/track/thriller> mo:performer <http://example.org/artist/michael_jackson>`

Here is an example of using Descriptions logics:

- $C_Major \sqsubseteq Scale$
- $G_Major_Triad \sqsubseteq Triad \sqcap \exists hasRoot.G \sqcap \exists hasChordType.Major$
- $C_Major \sqsubseteq \exists hasTriad.G_Major_Triad$
- $hasDominantTriad \equiv hasTriad \sqcap \exists hasDegree.5$
- $C_Major \sqsubseteq \exists hasDominantTriad.G_Major_Triad$

Symbolic Music

Music Graphs

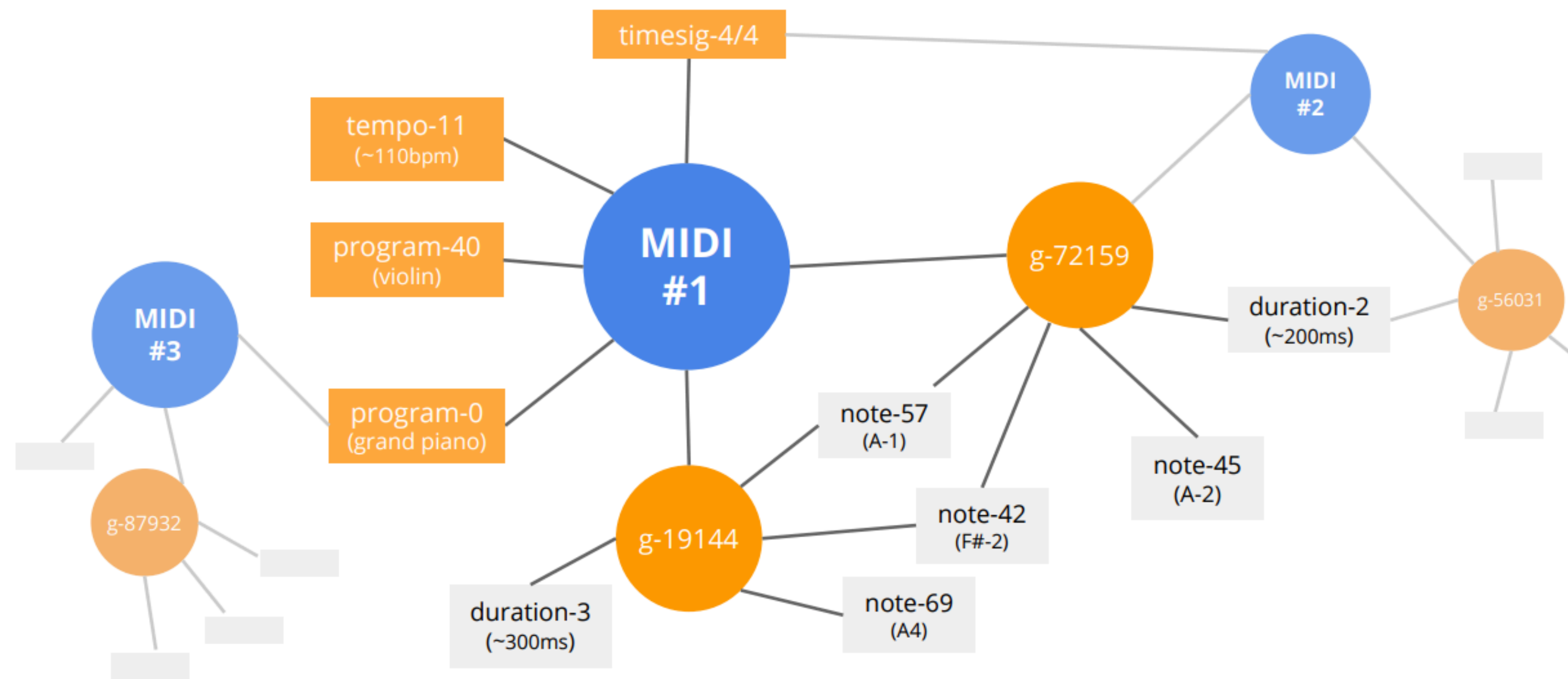


Graph Example from
The Music Meta Ontology

Carriero, Valentina, et al.
"The Music Meta Ontology: A flexible semantic model for the interoperability of music metadata." Proceedings of the 24th International Society for Music Information Retrieval Conference. 2023.

Symbolic Music

Music Graphs



Graph Example from
MIDI2Vec

Lisena, Pasquale, Albert Meroño-Peñuela, and Raphaël Troncy.
"MIDI2vec: Learning MIDI embeddings for reliable prediction
of symbolic music metadata." *Semantic Web* 13.3 (2022): 357-377.

Symbolic Music

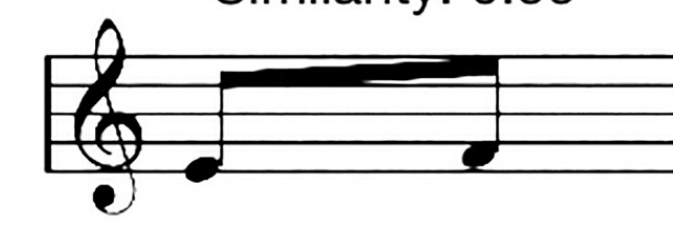
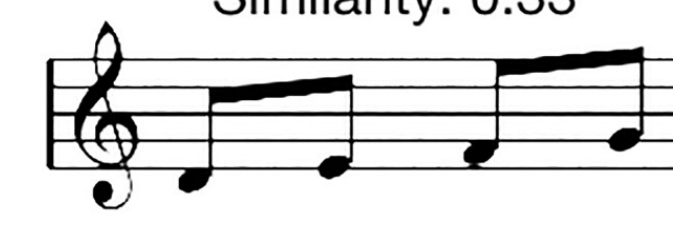
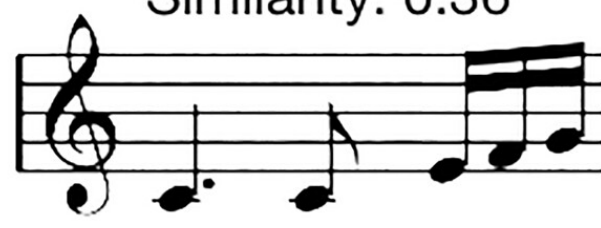
Transformer-based models for Symbolic Music Tasks

Model		Base model	MIR mechanism	Data	Representation	Tasks	Code
Encoder-only architecture							
<i>MTBert</i> [258]	(2023)	BERT (no pre-training)	–	4-part chorales	Interval + duration (event based)	Fugue form analysis	✗
Decoder-only architecture							
<i>Music Transformer</i> [99]	(2018)	Tf. decoder	Relative attention	Piano/Choral	MIDI-like	Priming/Harmonization	✓
<i>Pop Music Transformer</i> [101]	(2020)	Transformer-XL	–	Piano	REMI	Free generation/Priming	✓
<i>Jazz Transformer</i> [239]	(2020)	Transformer-XL	–	Lead sheet	REMI derived (Chords)	Free generation	✓
<i>PopMAG</i> [189]	(2020)	Transformer-XL	–	Multi-track	MuMIDI	Accompaniment generation	✗
Di et al. [48]	(2021)	Tf. decoder	–	Multi-track	CPWord derived (Rhythm family)	Video-to-music	✓
Chang et al. [18]	(2021)	XLNet	Relative bar encoding	Piano	Compound Word	Infilling	✓
<i>Compound Word Tf.</i> [94]	(2021)	Linear Tf. decoder	–	Piano	Compound Word	Free generation/Priming	✓
Sarmiento et al. [195]	(2021)	Transformer-XL	–	Guitar tabs + multi-track	DadaGP	Metadata-conditioned gen.	✓
<i>ComMU</i> [128]	(2022)	Transformer-XL	–	Multi-track	REMI + metadata	Metadata-conditioned gen. Multi-track combination	✓
<i>SymphonyNet</i> [144]	(2022)	Linear Tf.	3-D positional encoding	Orchestral	MMR	Free generation/Priming Chord-conditioned generation	✓
<i>Multitrack Music Tf.</i> [52]	(2023)	Tf. decoder	–	Orchestral	MMT	Free generation/Priming Instr.-conditioned generation	✓
<i>GTR-CTRL</i> [196]	(2023)	Transformer-XL	–	Guitar tabs + multi-track	DadaGP	Instr.-conditioned generation Genre-conditioned generation	✗
<i>Choir Transformer</i> [261]	(2023)	Tf. decoder	Relative attention	4-part chorales	Chord + pitch (event based)	Harmonization	✓
Guo et al. [80]	(2023)	Tf. encoder w/ custom attention	Fundamental music embedding RIPO attention	Monophonic	FME	Priming	✓

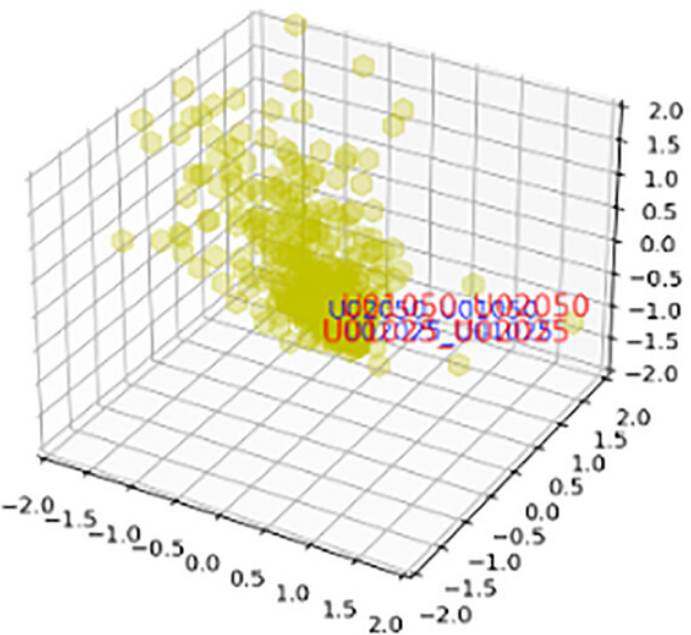
Dinh-Viet-Toan Le, Louis Bigo, Dorien Herremans, and Mikaela Keller. 2025. Natural Language Processing Methods for Symbolic Music Generation and Information Retrieval: A Survey. *ACM Comput. Surv.* 57, 7, Article 175 (July 2025), 40 pages. <https://doi.org/10.1145/3714457>

Symbolic Music

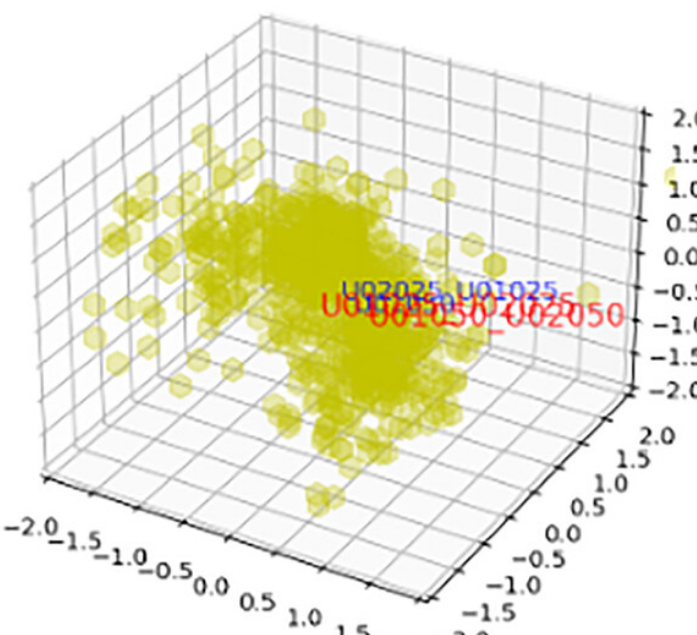
Pattern Discovery

 Example 1 : U01050_U02050	 Example 2 : U01025_U02025
 Most Similar (N=100) : U02050_U01050 Similarity: 0.55	 Most Similar (N=100) : U02025_U01025 Similarity: 0.43
 Most Similar (N=500) : U01050 Similarity: 0.33	 Most Similar (N=500) : U02025_U01025 Similarity: 0.36
 Most Similar (N=Full) : U02050_U01050_U02050 Similarity: 0.21	 Most Similar (N=Full) : E00150_U04100_U01025_U02025 Similarity: 0.21

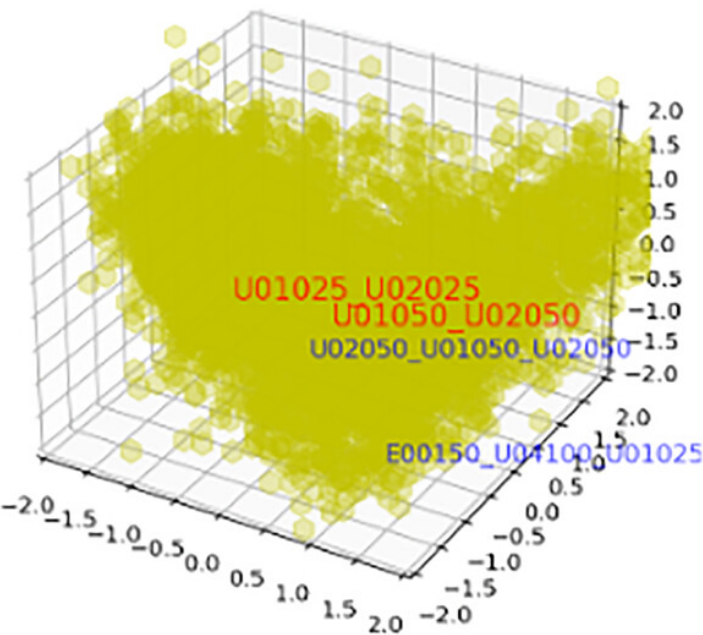
PCA Visualization (N = 100)



PCA Visualization (N = 500)



PCA Visualization (N = Full)



Examples of finding the most similar **Mel2Word** based on word2vec vectorisation.

Park, S., Choi, E., Kim, J., & Nam, J. (2024).
Mel2Word: A Text-Based Melody Representation
for Symbolic Music Analysis.
Music & Science, 7, 20592043231216254.

Symbolic Music

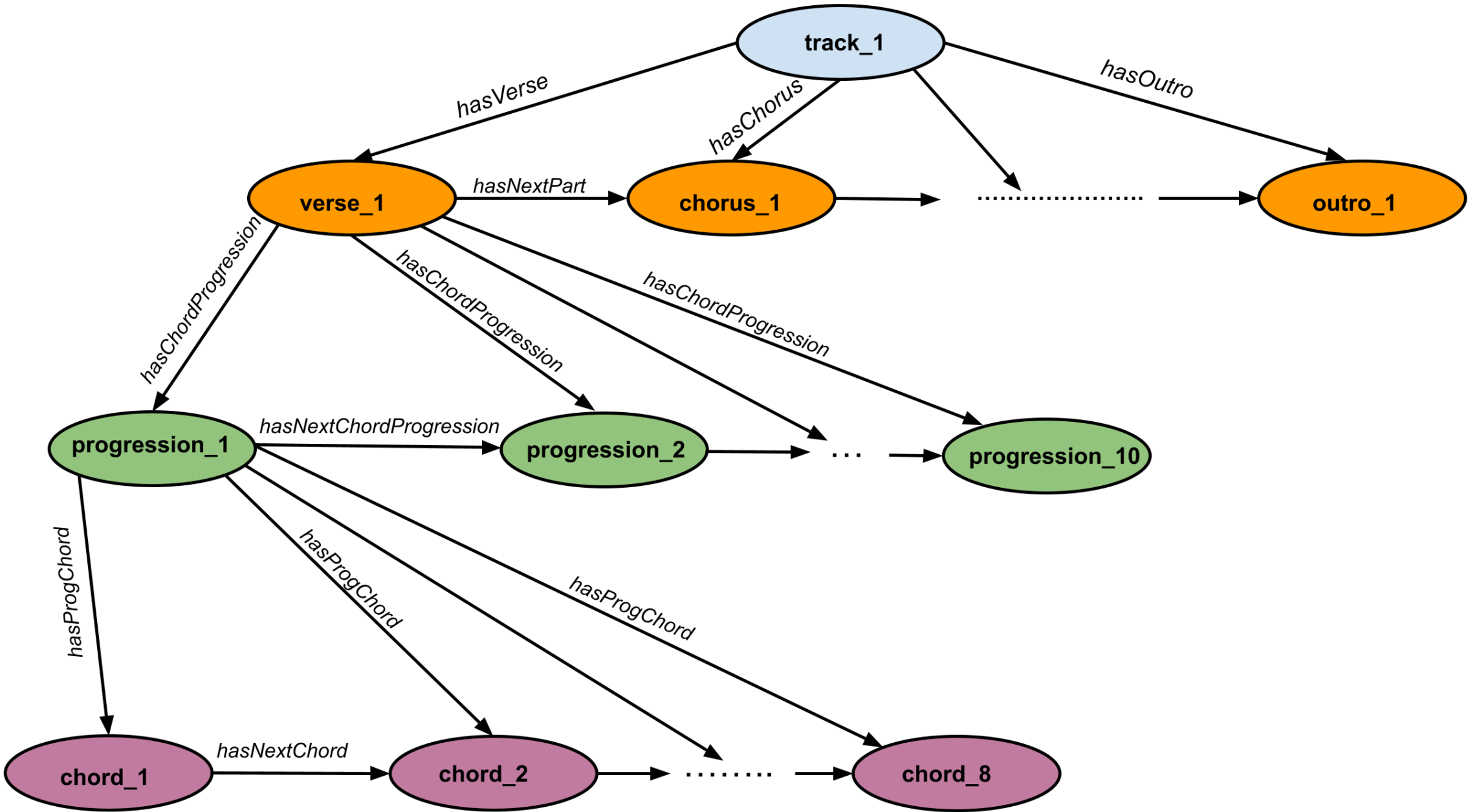
Structural, Sequential and Harmonic Analysis

Tonic $\sqcap \exists \text{hasPrevious.ParallelDominant} \sqsubseteq \text{LocalTonic}$

DominantChord $\sqcap \exists \text{hasNext.LocalTonic} \sqsubseteq \text{LocalDominant}$

DominantChord $\sqcap \exists \text{hasNext.LocalDominant} \sqsubseteq \text{LocalSubdominant}$

Chord	Inferred Types	
Am7b5	LocalSubdominant	D_PhrygianParallelDominant
D7	LocalDominant	G_IonianParallelDominant
Gm7	G_AeolianSongTonic	C_AeolianParallelDominant
Cm7	LocalSubdominant	F_MixolydianParallelDominant
F7	LocalDominant	B_ParallelDominant
BbΔ	G_AeolianRelativeTonic	E_LydianParallelDominant
EbΔ	LocalTonic	A_LocrianParallelDominant
Am7b5	LocalSubdominant	D_ParallelDominant
D7	LocalDominant	G_ParallelDominant
Gm7	G_AeolianSongTonic	C_MixolydianDominant
C9	LocalDominant	F_ParallelDominant
Fm7	LocalTonic	Bb_AeolianDominant
B7	LocalSubdominant	Eb_ParallelDominant
Eb7	LocalDominant	A_LocrianDominant
Am7b5	LocalTonic	-
D7b13	LocalDominant	G_ParallelDominant
Gm	G_AeolianSongTonic	-



Kantarelis, S., Dervakos, E., Kotsani, N., & Stamou, G. (2023). Functional harmony ontology: Musical harmony analysis with Description Logics. Journal of Web Semantics, 75, 100754.

Kantarelis, S., Dervakos, E., & Stamou, G. (2023). The Music Part Ontology. In Description Logics.

Symbolic Music

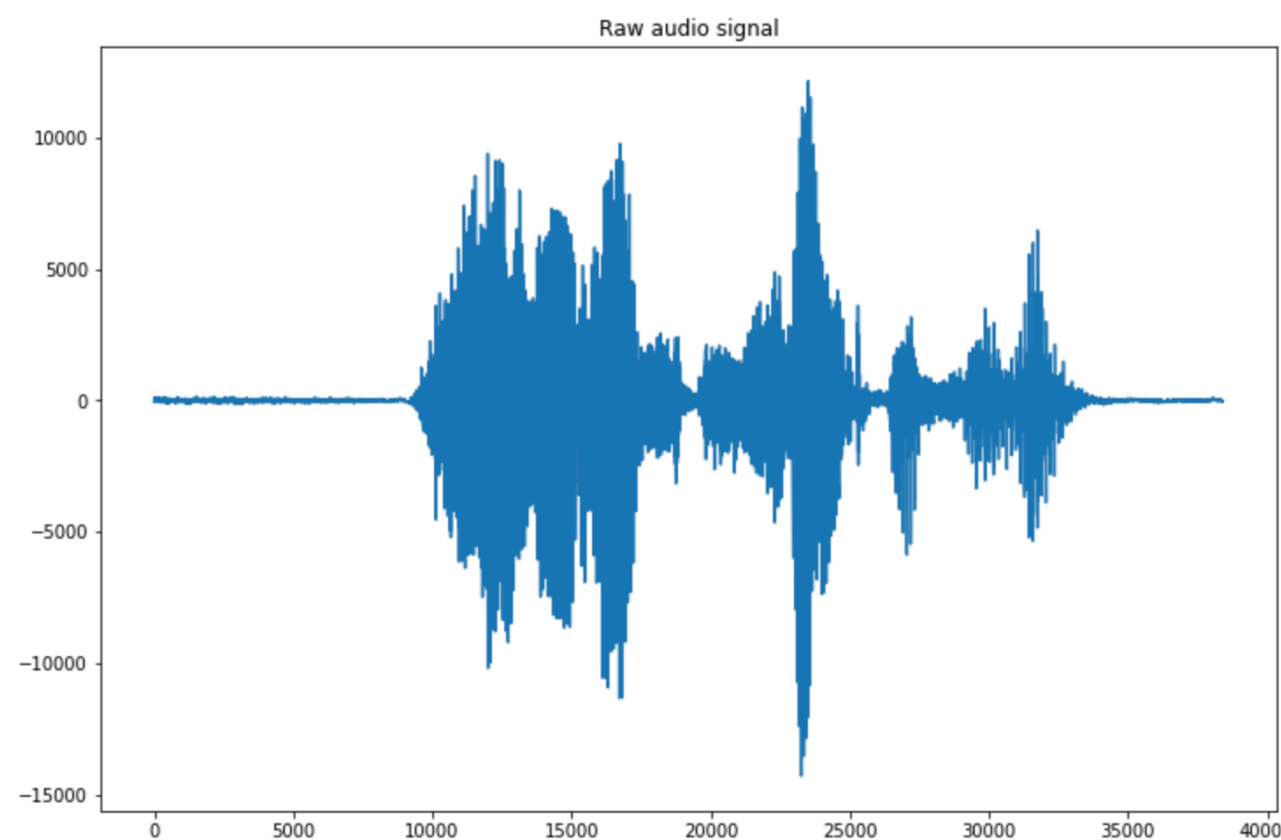
Advantages and Limitations

- Text-like format → NLP methods applicable
- Lightweight → no need for deep models
- Access to basic elements (e.g., notes/chords) → ideal for music analysis
- No standardization → high data curation cost
- Poor audio quality (e.g., MIDI) → not ideal for generation
- Few high-quality datasets

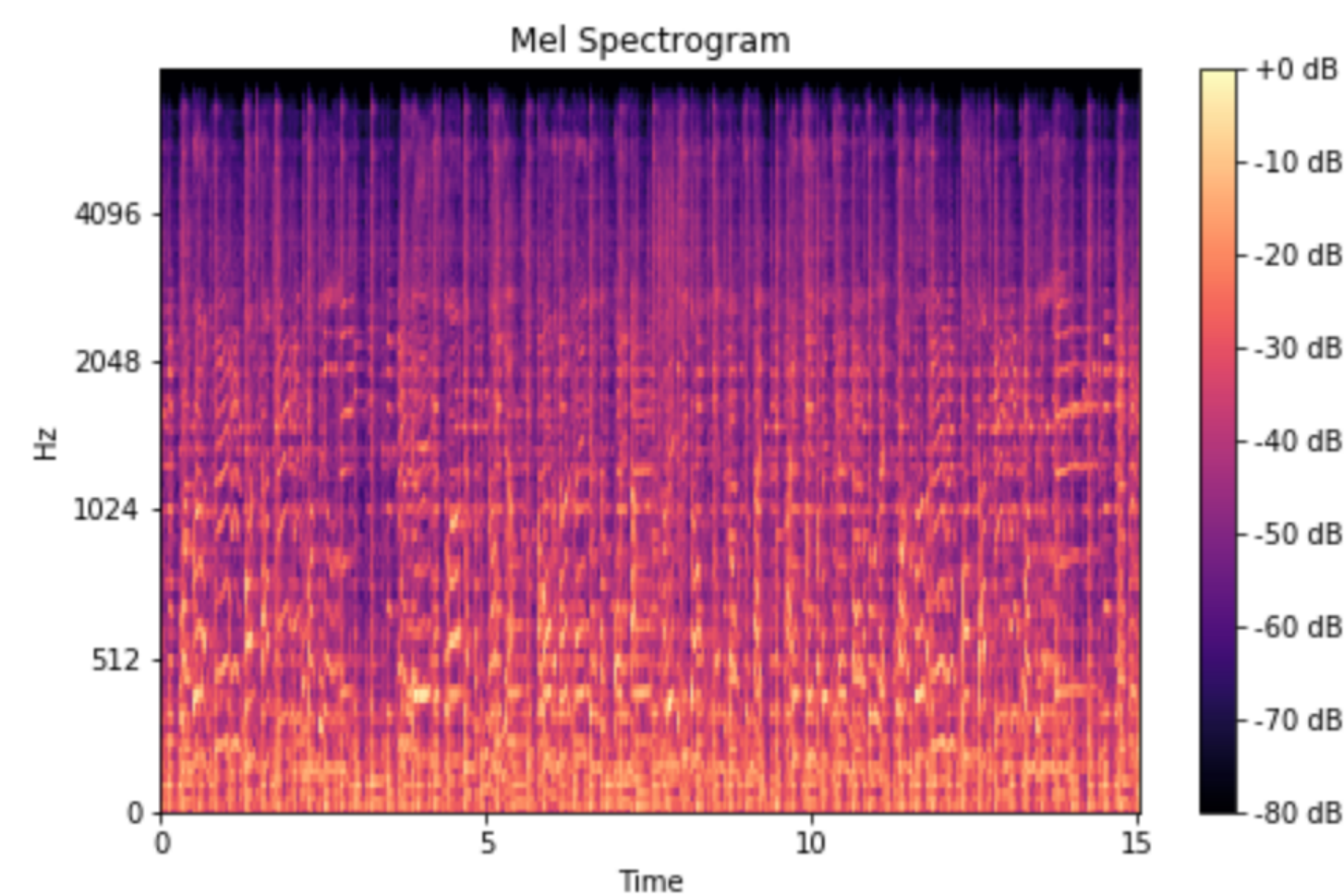
Audio

Audio Representation

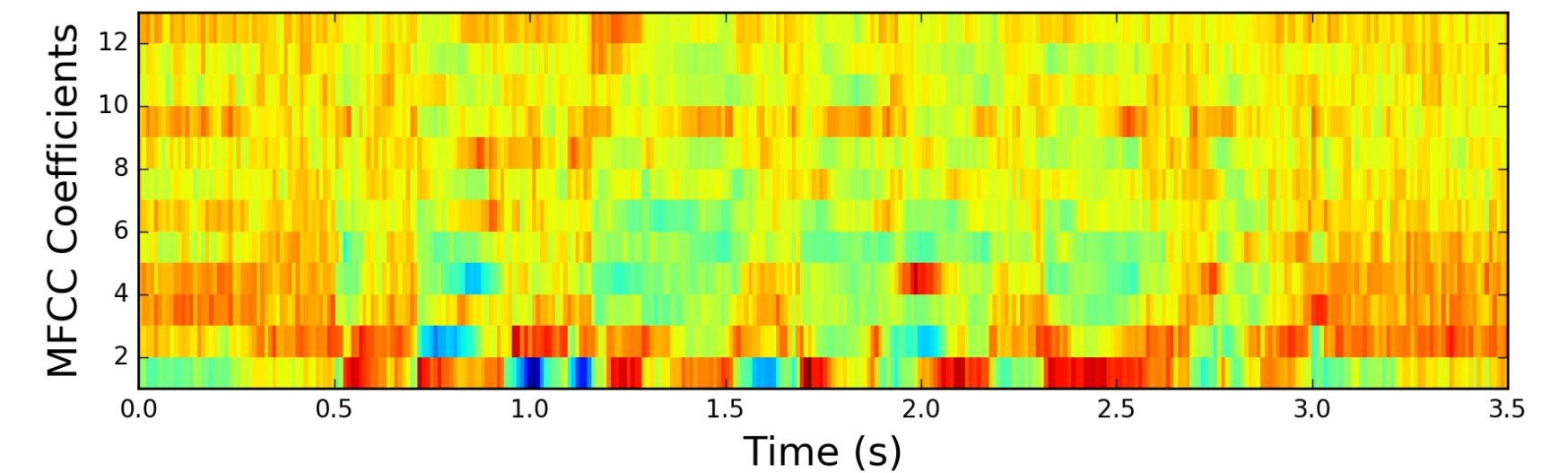
Audio provides detailed **frequency-based** features, supporting analysis of **sound characteristics** like timbre, dynamics, and expressive performance.



Raw Audio



Mel Spectrogram



Mel-frequency cepstral coefficients
(MFFCCs)

Audio

Mel-Spectrograms vs MFCCs

Mel-Spectrogram

- Time–frequency representation
- Y-axis: Mel-frequency bins (human ear-like scale)
- Retains fine spectral details
- High-dimensional, detailed (good for deep learning)
- Common in audio classification, sound event detection

MFCCs

- Compact spectral representation
- Y-axis: Mel-frequency cepstral coefficients
- Focuses on overall spectral shape
- Low-dimensional, good for traditional ML
- Used in speech recognition, audio feature extraction

Audio

Basic Tools

- Signal Processing Tools
 - Essential - <https://essentia.upf.edu/>
 - Librosa (python library) - <https://librosa.org/doc/latest/index.html>
- audioFX (python library) - <https://pypi.org/project/audioFX/>
- Spotify API - <https://developer.spotify.com/documentation/web-api>
- DNNs
- Transformers
- Diffusion Models

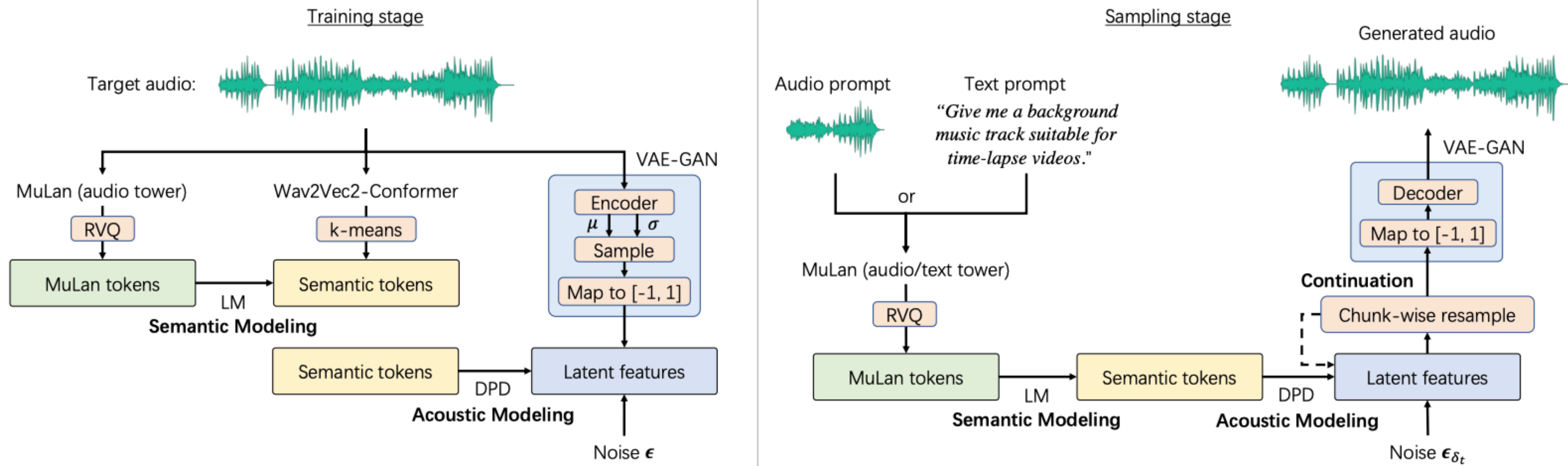
Audio

Main Tasks

- Music Generation
- Feature Extraction and Detection
- Audio Classification
- Auto-tagging
- Emotion related tasks
- Expression related tasks
- Source Separation
- Music Understanding

Audio

Audio Generation

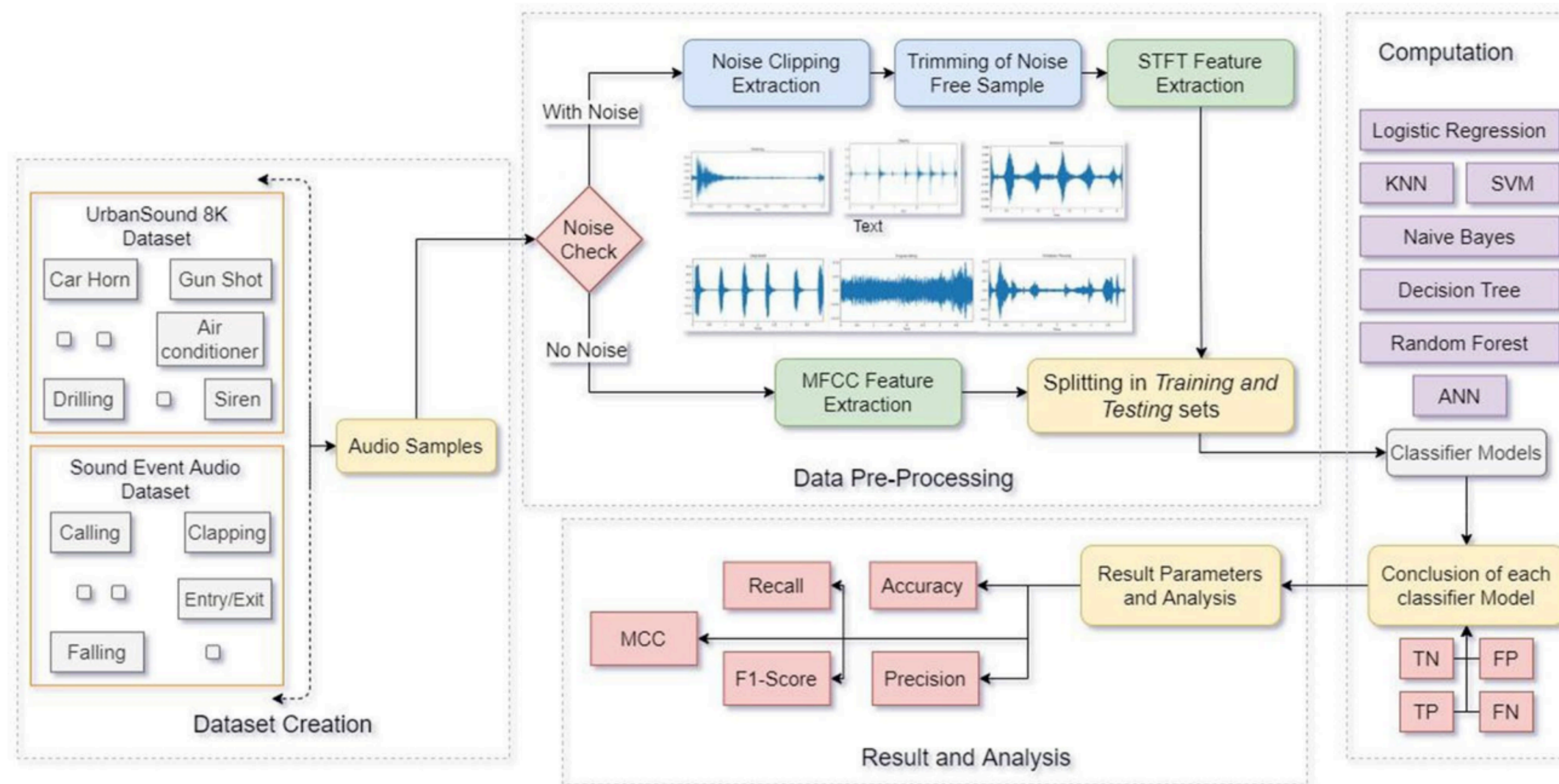


The training and sampling pipelines of MeLoDy

Lam, Max WY, et al. "Efficient neural music generation."
Advances in Neural Information Processing Systems 36 (2023): 17450-17463.

Audio

Audio Classification

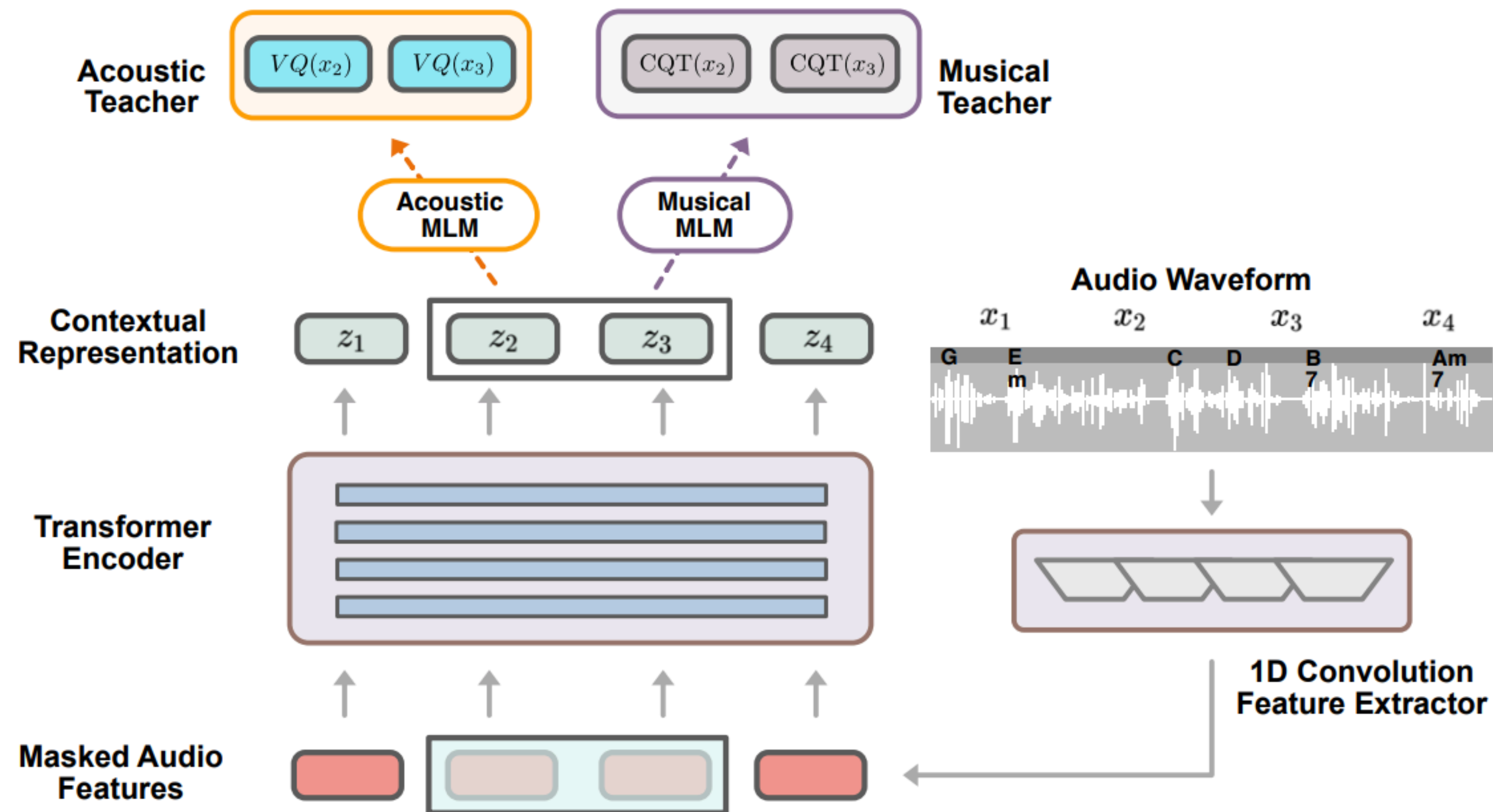


Gourisaria, Mahendra Kumar, et al.
 "Comparative analysis of audio classification with MFCC and STFT features using machine learning techniques."
 Discover Internet of Things 4.1 (2024): 1.

Flowchart of various steps followed for classification of audio samples

Audio

Music Understanding



Yizhi, L. I., et al. "MERT: Acoustic music understanding model with large-scale self-supervised training." The Twelfth International Conference on Learning Representations. 2023.

Illustration of the **MERT** Pre-training Framework.

Audio

Music Understanding

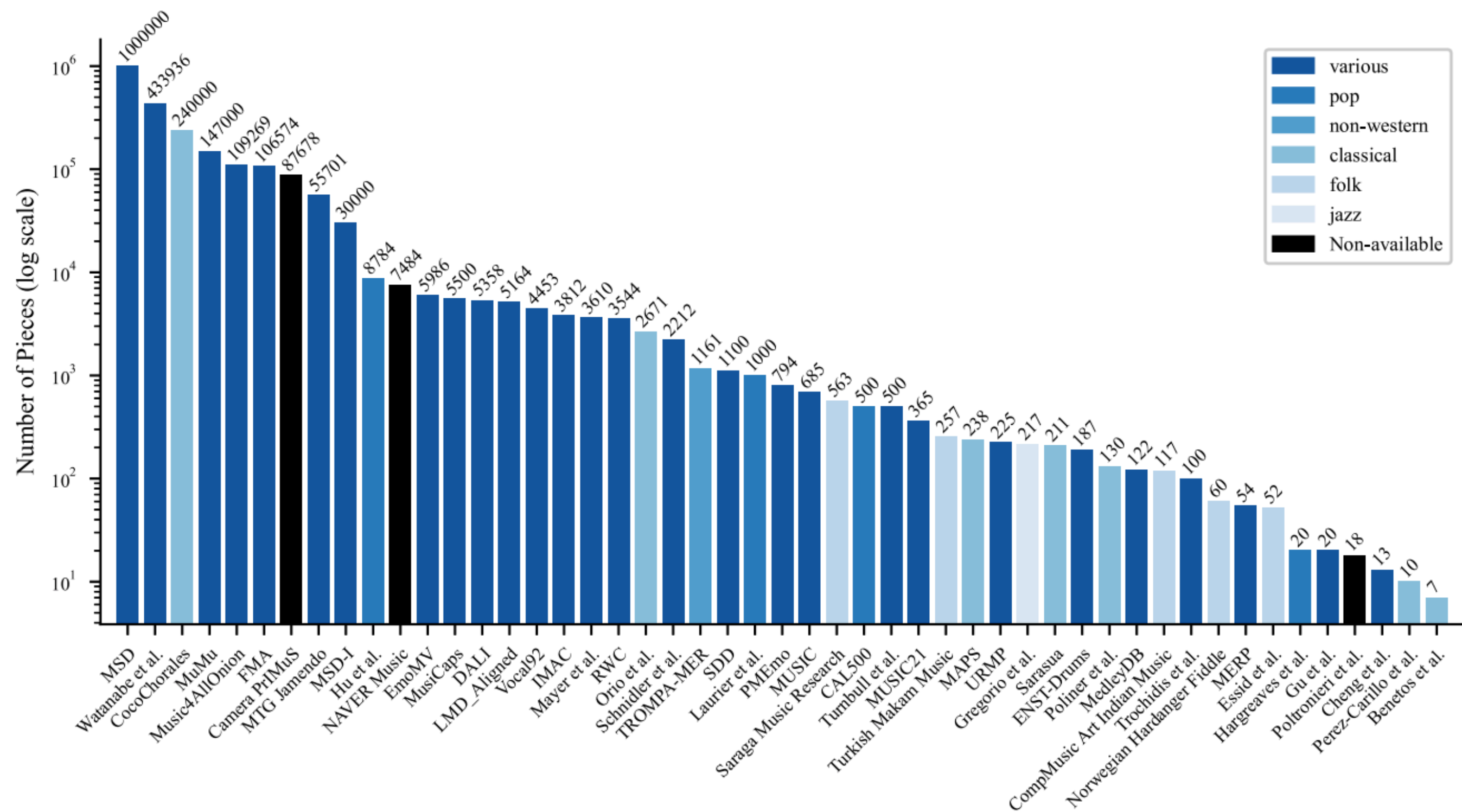
Dataset Task	MTT Tagging		GS Key	GTZAN Genre	GTZAN Rhythm	EMO Emotion		Nsynth Instrument	Pitch	VocalSet Tech	VocalSet Singer
Metrics	ROC	AP	Acc ^{Refined}	Acc	F1 ^{beat}	R2 ^V	R2 ^A	Acc	Acc	Acc	Acc
MusiCNN [41]	90.6*	38.3*	12.8*	79.0*	-	46.6*	70.3*	72.6	64.1	70.3	57.0
CLMR [48]	89.4*	36.1*	14.9*	68.6*	-	45.8*	67.8*	67.9	47.0	58.1	49.9
Jukebox-5B [15; 57]	91.5*	41.4*	66.7*	79.7*	-	61.7*	72.1*	70.4	91.6	76.7	82.6
MULE [36]	91.4*	40.4*	66.7*	73.5*	-	57.7*	70.0*	74.0*	89.2*	75.5	87.5
HuBERT-base ^{music} [25]	90.2	37.7	14.7	70.0	88.6	42.1	66.5	69.3	77.4	65.9	75.3
data2vec-base ^{music} [2]	90.0	36.2	50.6	74.1	68.2	52.1	71.0	69.4	93.1	71.1	81.4
MERT-95M ^{K-means}	90.6	38.4	65.0	78.6	88.3	52.9	69.9	71.3	92.3	74.6	77.2
MERT-95M-public ^{K-means}	90.7	38.4	67.3	72.8	88.1	59.7	72.5	70.4	92.3	75.6	78.0
MERT-95M ^{RVQ-VAE}	91.0	39.3	63.5	78.6	88.3	60.0	76.4	70.7	92.6	74.2	83.7
MERT-330M ^{RVQ-VAE}	91.3	40.2	65.6	79.3	87.9	61.2	74.7	72.6	94.4	76.9	87.1
(Previous) SOTA	92.0 [26]	41.4 [15]	74.3 [30]	83.5 [36]	80.6 [24]	61.7	72.1 [15]	78.2 [53]	89.2 [36]	65.6 [55]	80.3 [39]

Experimental Performance of **MERT** and Baselines on Downstream Tasks

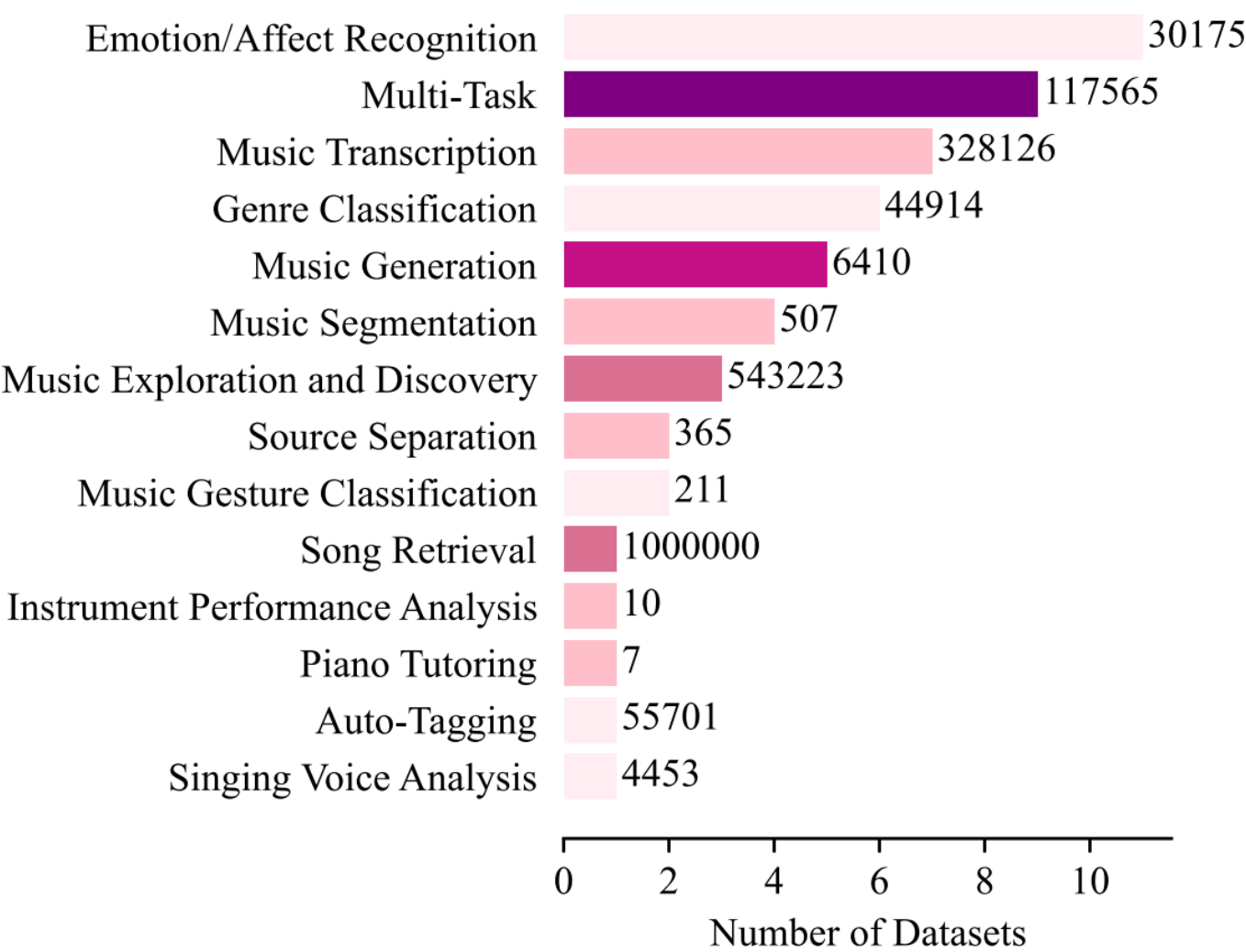
Yizhi, L. I., et al. "MERT: Acoustic music understanding model with large-scale self-supervised training." The Twelfth International Conference on Learning Representations. 2023.

Audio

Fundamental Datasets



Distribution of dataset sizes across various music genres



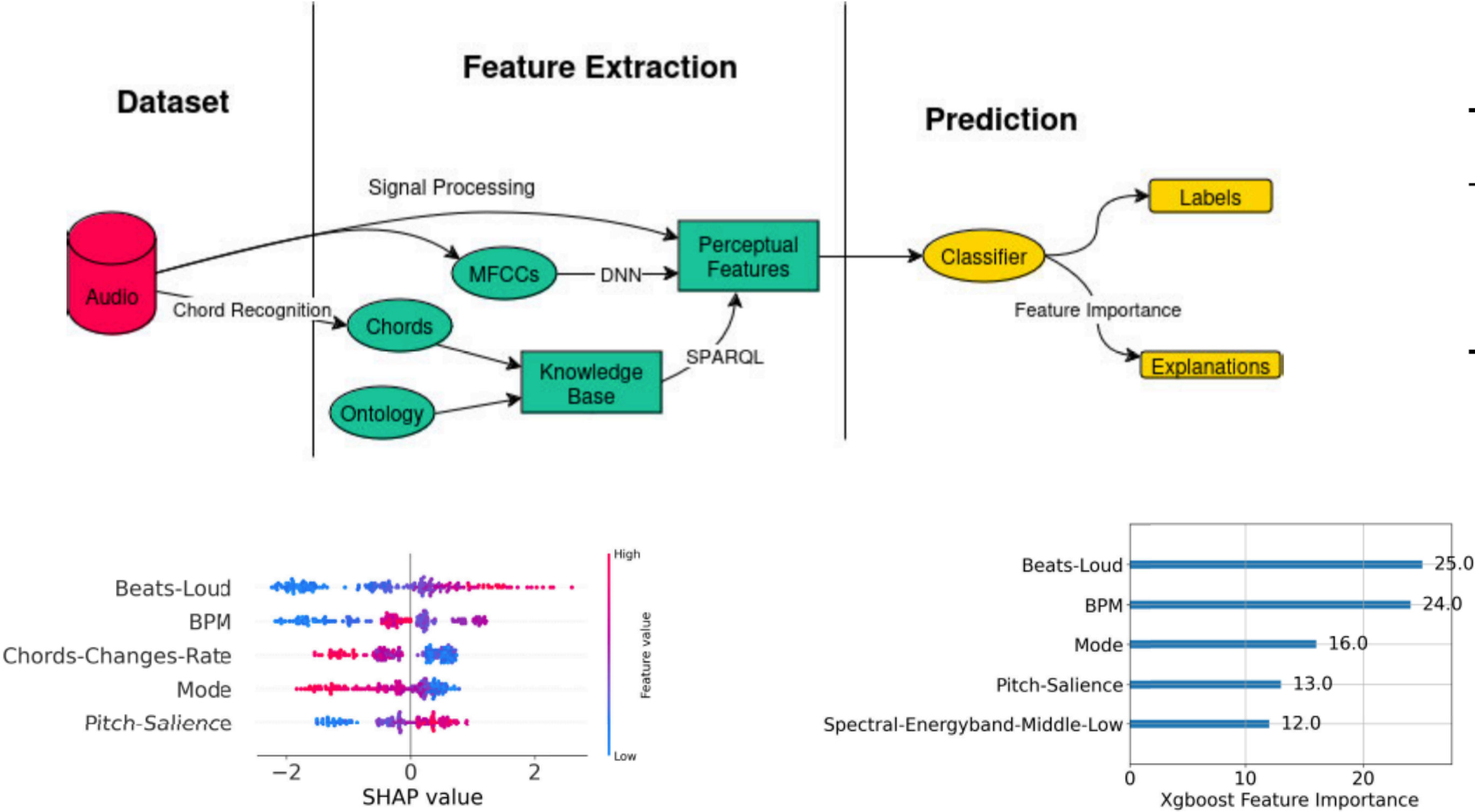
Dataset distribution across music processing tasks

Christodoulou, A. M., Lartillot, O., & Jensenius, A. R. (2024). Multimodal music datasets? Challenges and future goals in music processing. International Journal of Multimedia Information Retrieval



Audio

Interpretable Music Auto-Tagging



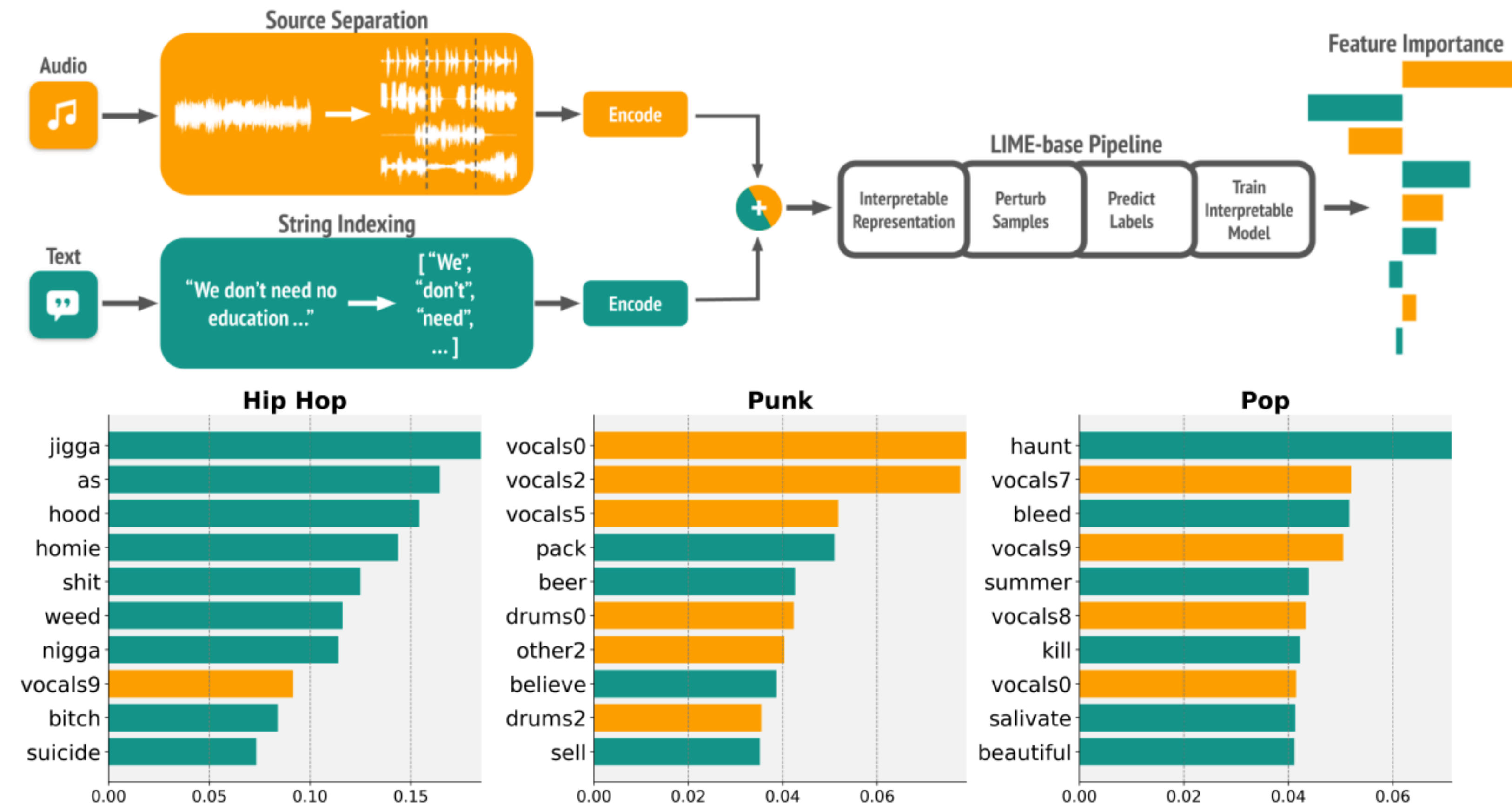
Explanations for *Disco* label

Dataset	Metric	XGBoost	SOTA
MTG Jamendo	ROC-AUC	0.72	0.76 [6]
MagnaTagATune	ROC-AUC	0.84	0.91 [16]
GTZAN	Accuracy	0.79	0.84 [50]

Lyberatos, V., Kantarelis, S., Dervakos, E., & Stamou, G. (2025). Challenges and Perspectives in Interpretable Music Auto-Tagging using Perceptual Features. IEEE Access.

Audio

Explainable Multimodal Music Understanding



Sotirou, T., Lyberatos, V., Mastromichalakis, O. M., & Stamou, G. (2025, April). Musiclime: Explainable multimodal music understanding. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 1-5). IEEE.

Audio

Advantages and Limitations

- Rich, realistic sound representation
- Includes performance nuances (dynamics, articulation)
- Compatible with **human perception** (frequencies, timbre)
- Suitable for generation, synthesis, style transfer
- Broad dataset availability
- High-dimensional, storage-heavy
- Needs **deep models** for most tasks
- Harder to extract symbolic elements (notes, chords)
- Sensitive to noise and recording conditions
- **Lower interpretability** for analysis

Music and AI

Limitations and Challenges

- Few large, high-quality, diverse datasets
- Fragmented formats (audio, symbolic, scores)
- Most datasets use copyright-free or synthetic music
- No datasets reflect real-world popular music consumption
- Expressiveness and timing nuance difficult to capture
- No standard evaluation metrics for creativity or quality
- Human evaluations costly, subjective
- Lack of agreed benchmarks across tasks
- Emotion, genre and mood perception is subjective and culturally influenced

Music and AI

Ethics

- AI mimics real artists → Deepfake risks
- Trained on copyrighted music → Legal ambiguity, ownership disputes
- Mass AI-generated music → Job loss for musicians, creative devaluation (?)
- Hidden training data → Lack of transparency, ethical opacity
- Indistinguishable human/AI output → Trust issues, listener deception

Music and AI

References

- 1.Carriero, Valentina, et al. "The Music Meta Ontology: A flexible semantic model for the interoperability of music metadata." Proceedings of the 24th International Society for Music Information Retrieval Conference. 2023.
- 2.Lisena, Pasquale, Albert Meroño-Peñuela, and Raphaël Troncy. "MIDI2vec: Learning MIDI embeddings for reliable prediction of symbolic music metadata." Semantic Web 13.3 (2022): 357-377.
- 3.Dinh-Viet-Toan Le, Louis Bigo, Dorien Herremans, and Mikaela Keller. "Natural Language Processing Methods for Symbolic Music Generation and Information Retrieval: A Survey." ACM Comput. Surv. 57, 7, Article 175 (July 2025), 40 pages. <https://doi.org/10.1145/3714457>
- 4.Park, S., Choi, E., Kim, J., & Nam, J. (2024). "Mel2Word: A Text-Based Melody Representation for Symbolic Music Analysis." Music & Science, 7, 20592043231216254.
- 5.Kantarelis, S., Dervakos, E., & Stamou, G. (2023). "The Music Part Ontology." In Description Logics.
- 6.Kantarelis, S., Dervakos, E., Kotsani, N., & Stamou, G. (2023). "Functional harmony ontology: Musical harmony analysis with Description Logics." Journal of Web Semantics, 75, 100754.
- 7.Lam, Max WY, et al. "Efficient neural music generation." Advances in Neural Information Processing Systems 36 (2023): 17450-17463.
- 8.Gourisaria, Mahendra Kumar, et al. "Comparative analysis of audio classification with MFCC and STFT features using machine learning techniques." Discover Internet of Things 4.1 (2024): 1.
- 9.Yizhi, L. I., et al. "MERT: Acoustic music understanding model with large-scale self-supervised training." The Twelfth International Conference on Learning Representations. 2023.
- 10.Christodoulou, A. M., Lartillot, O., & Jensenius, A. R. (2024). "Multimodal music datasets? Challenges and future goals in music processing." International Journal of Multimedia Information Retrieval.
- 11.Lyberatos, V., Kantarelis, S., Dervakos, E., & Stamou, G. (2025). "Challenges and Perspectives in Interpretable Music Auto-Tagging using Perceptual Features." IEEE Access.
- 12.Sotirou, T., Lyberatos, V., Mastromichalakis, O. M., & Stamou, G. (2025, April). "Musiclime: Explainable multimodal music understanding." In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.

Music and AI

Recommended Handbooks and Special Issues

Handbook of Artificial Intelligence for Music
Foundations, Advanced Approaches, and Developments for Creativity
<https://link.springer.com/book/10.1007/978-3-030-72116-9>

Fundamentals of Music Processing
Using Python and Jupyter Notebooks
<https://link.springer.com/book/10.1007/978-3-030-69808-9>

Transactions of the International Society for Music Information Retrieval
<https://transactions.ismir.net/issue/archive>

Music and AI

Thank you

Spyros Kantarelis, PhD student at AILS Lab, ECE NTUA

spyroskanta@ails.ece.ntua.gr